Lahore Urban Estates Analytics Warehouse

Fact and Dimension Identification

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This document identifies LUE's business processes, chooses a fact table type for each, identifies the conformed dimensions with their hierarchies and SCD policy, and defines the key strategy. It follows the four-step dimensional design process from Lab 05: business process, then grain, then dimensions, then facts.

1. Business Processes and Fact Table Types

P1. Rent invoicing and payment

Fact table(s): fact_rent_invoice_line, fact_payment
Fact type: Transaction (two facts)
Why this type: Billing and paying are separate events with different grains: one line can be paid in several instalments and one payment can cover several lines.

P2. Unit occupancy

Fact table(s): fact_unit_occupancy_daily
Fact type: Periodic snapshot
Why this type: Management needs the state of every unit on every day, including days on which nothing happened.

P3. Maintenance

Fact table(s): fact_maintenance_request
Fact type: Accumulating snapshot
Why this type: A request has a fixed set of milestones; one row per request with a date per milestone gives lags directly.

P4. Lease lifecycle

Fact table(s): fact_lease_event
Fact type: Factless (event tracking)
Why this type: The event itself is the fact; counting events by type answers renewal and churn questions.

P5. Vacancy marketing

Fact table(s): fact_vacancy_incentive_coverage
Fact type: Factless (coverage)
Why this type: Records offers that existed even when no lease followed, which no transaction fact can show.

The five processes are never combined into one fact table, because each has a different grain (Lab 05 Task 3.3). The grain sentences are defined in the Grain Definition document.

2. Fact Tables

fact_rent_invoice_line

Type: Transaction
Measures: billed_amount, tax_amount, discount_amount, net_amount
Degenerate dimensions: invoice_number, invoice_line_no
Expected rows: ≈ 130,000

fact_payment

Type: Transaction
Measures: payment_amount, days_past_due, settlement_lag_days
Degenerate dimensions: payment_reference, invoice_number, invoice_line_no
Expected rows: ≈ 60,000

fact_unit_occupancy_daily

Type: Periodic snapshot
Measures: is_occupied, contract_rent, current_lease_arrears
Degenerate dimensions: none
Expected rows: ≈ 1.97 million

fact_maintenance_request

Type: Accumulating snapshot
Measures: days_to_assign, days_to_resolve, days_to_verify, estimated_cost, actual_cost, sla_breached_flag
Degenerate dimensions: request_number, request_status
Expected rows: ≈ 6,000

fact_lease_event

Type: Factless (event tracking)
Measures: event_count
Degenerate dimensions: none
Expected rows: ≈ 8,000

fact_vacancy_incentive_coverage

Type: Factless (coverage)
Measures: none (factless)
Degenerate dimensions: none
Expected rows: Set in Phase 2 (depends on generator settings)

3. Bus Matrix

Rows are dimensions and columns are fact tables. An X marks a dimension used by a fact; for dim_date the date roles are listed instead. A dimension with several marks is conformed: the same table, keys and attribute values serve every process, which is what allows drill-across questions such as Q12.

Dimension	rent_invoice_line	payment	unit_occupancy_daily	maintenance_request	lease_event	vacancy_incentive_coverage
dim_date	invoice, due	paid, due	snapshot	raised, assigned, resolved, verified	event	coverage
dim_unit	X	X	X	X	X	X
dim_tenant	X	X	X	X	X	
dim_lease	X	X	X		X	
dim_owner	X	X	X	X	X	X
dim_payment_method		X				
dim_charge_type	X	X				
dim_maintenance_category				X		
dim_vendor				X		
dim_lease_event_type					X	
dim_request_flags				X		
dim_incentive						X
dim_weather				X		

4. Dimensions

dim_date

One row per: calendar day from 2023-07-01 to 2027-06-30
Natural key: full_date
Surrogate key: date_key
Attributes: full_date, day_of_month, day_name, is_weekend, week_of_year, month_no, month_name, quarter_no, year_no, fiscal_month_no, fiscal_quarter_no, fiscal_year_label, is_public_holiday, holiday_name, is_ramadan
Hierarchy: Day → Month → Quarter → Year; Day → Fiscal month → Fiscal quarter → Fiscal year
Roles: invoice, due, paid, snapshot, raised, assigned, resolved, verified, event, coverage

dim_unit

One row per: unit version (SCD Type 2)
Natural key: unit_nk (building_code + unit_no)
Surrogate key: unit_key
Attributes: unit_nk, building_code, unit_no, unit_type, bedrooms, area_sqft, furnished_flag, floor_no, building_name, building_class, building_age_band, locality, city, region
Hierarchy: Unit → Floor → Building → Locality → City → Region

dim_tenant

One row per: tenant version (SCD Type 2)
Natural key: tenant_code
Surrogate key: tenant_key
Attributes: tenant_code, tenant_name, tenant_type, employer_sector, contact_channel, is_inferred
Hierarchy: Tenant type → Tenant

dim_lease

One row per: lease contract
Natural key: lease_number
Surrogate key: lease_key
Attributes: lease_number, lease_type, start_date, end_date, term_months, original_monthly_rent, rent_review_type, previous_lease_number
Hierarchy: —

dim_owner

One row per: owner
Natural key: owner_code
Surrogate key: owner_key
Attributes: owner_code, owner_name, owner_type, portfolio_size_band
Hierarchy: Owner type → Owner

dim_payment_method

One row per: payment method
Natural key: method_code
Surrogate key: payment_method_key
Attributes: method_code, method_name, channel_group, typical_settlement_days
Hierarchy: Method → Channel group

dim_charge_type

One row per: charge type
Natural key: charge_code
Surrogate key: charge_type_key
Attributes: charge_code, charge_type_name, charge_category
Hierarchy: Charge type → Charge category

dim_maintenance_category

One row per: maintenance subcategory version (SCD Type 2)
Natural key: subcategory_code
Surrogate key: maint_category_key
Attributes: subcategory_code, subcategory_name, category_name, default_sla_hours
Hierarchy: Subcategory → Category

dim_vendor

One row per: vendor version (SCD Type 2)
Natural key: vendor_code
Surrogate key: vendor_key
Attributes: vendor_code, vendor_name, trade, rating_band, phone
Hierarchy: Vendor → Trade

dim_lease_event_type

One row per: lease event type
Natural key: event_code
Surrogate key: lease_event_type_key
Attributes: event_code, event_name, is_exit_event
Hierarchy: —

dim_request_flags

One row per: combination of the three request flags (junk dimension, 8 rows)
Natural key: is_emergency + is_recurring + is_tenant_caused
Surrogate key: request_flags_key
Attributes: is_emergency, is_recurring, is_tenant_caused, flags_label
Hierarchy: —

dim_incentive

One row per: incentive offer type
Natural key: incentive_code
Surrogate key: incentive_key
Attributes: incentive_code, incentive_type, incentive_description, discount_pct, rent_free_days
Hierarchy: Incentive type → Incentive

dim_weather

One row per: city per calendar day
Natural key: city + weather_date
Surrogate key: weather_key
Attributes: city, weather_date, rainfall_mm, rain_band, max_temp_c, heat_band
Hierarchy: —

5. Hierarchies

Hierarchies are flattened into their dimension (a star, not a snowflake), as required by Lab 04's design rule. Drill-down moves from the lowest level to the highest; roll-up moves the other way.

- **dim_date** (Calendar): Day → Month → Quarter → Year
- **dim_date** (Fiscal (July–June)): Day → Fiscal month → Fiscal quarter → Fiscal year
- **dim_unit** (Location): Unit → Floor → Building → Locality → City → Region
- **dim_payment_method** (Channel): Method → Channel group
- **dim_charge_type** (Charge): Charge type → Charge category
- **dim_maintenance_category** (Maintenance): Subcategory → Category
- **dim_vendor** (Trade): Vendor → Trade
- **dim_tenant** (Tenant type): Tenant → Tenant type
- **dim_owner** (Owner type): Owner → Owner type
- **dim_incentive** (Incentive): Incentive → Incentive type

6. SCD Policy

Every descriptive attribute has a declared type. *Type 0* attributes never change. *Type 1* attributes are corrected in place for all history. *Type 2* attributes create a new dimension row with effective_from, effective_to and is_current, so facts keep pointing at the version valid at the time of the event.

dim_date

Type 1: is_public_holiday (holiday dates are corrected when officially announced)
Type 0: all calendar and fiscal attributes (a date's attributes never change)

dim_unit

Type 2: unit_type (a converted unit must keep past rent and occupancy under its old type); furnished_flag (furnishing changes the rent level; history needed for rent analysis)
Type 1: bedrooms (corrections only); area_sqft (corrections only); building_name (renames and spelling fixes apply to all history)
Type 0: unit_nk (identity of the unit); building_code (fixed by definition or contract); unit_no (fixed by definition or contract); floor_no (fixed by definition or contract)

dim_tenant

Type 2: tenant_type (Q4 compares late payment by tenant type at the time of payment); employer_sector (Q4 analyses late payment by employer sector over time)
Type 1: tenant_name (spelling corrections; no analytical history); contact_channel (current preference only)
Type 0: tenant_code (identity)

dim_lease

Type 0: lease_number (identity); lease_type (signed contract term); start_date (signed contract term); end_date (signed contract term; a renewal is a new lease); term_months (fixed by definition or contract); original_monthly_rent (signed contract term); rent_review_type (fixed by definition or contract); previous_lease_number (fixed by definition or contract)

dim_owner

Type 1: owner_name (corrections only); portfolio_size_band (current band only; ownership history lives on the facts)
Type 0: owner_code (fixed by definition or contract)

dim_maintenance_category

Type 2: default_sla_hours (a tighter sla policy must not re-label past requests as breaches)

Type 0: subcategory_code (fixed by definition or contract)

dim_vendor

Type 2: rating_band (Q9 asks whether vendors were slow while rated low)

Type 1: phone (contact fix)

Type 0: vendor_code (fixed by definition or contract)

dim_weather

Type 1: rainfall_mm (provider revisions overwrite the value)

Attributes not listed (for example method names and flag labels) are Type 1. The complete list is in the data dictionary, sheet SCD_Policy.

7. Keys

Natural and surrogate keys

- **dim_date**: natural key full_date → surrogate key date_key
- **dim_unit**: natural key unit_nk (building_code + unit_no) → surrogate key unit_key
- **dim_tenant**: natural key tenant_code → surrogate key tenant_key
- **dim_lease**: natural key lease_number → surrogate key lease_key
- **dim_owner**: natural key owner_code → surrogate key owner_key
- **dim_payment_method**: natural key method_code → surrogate key payment_method_key
- **dim_charge_type**: natural key charge_code → surrogate key charge_type_key
- **dim_maintenance_category**: natural key subcategory_code → surrogate key maint_category_key
- **dim_vendor**: natural key vendor_code → surrogate key vendor_key
- **dim_lease_event_type**: natural key event_code → surrogate key lease_event_type_key
- **dim_request_flags**: natural key is_emergency + is_recurring + is_tenant_caused → surrogate key request_flags_key
- **dim_incentive**: natural key incentive_code → surrogate key incentive_key
- **dim_weather**: natural key city + weather_date → surrogate key weather_key

Why surrogate keys even when natural keys are unique (Lab 05 Task 7.2)

1. **History.** An SCD Type 2 tenant has several rows for one tenant_code; only a surrogate key can point at one version.
2. **Independence from source systems.** If PropertyHub renumbers tenants or LUE changes systems, warehouse keys and history are unaffected.
3. **Multi-source integration.** Unit A-101 exists in several buildings in the S6 spreadsheets, so the natural key becomes building_code + unit_no and facts join on unit_key (Lab 05 Scenario D).
4. **Special members and performance.** Integer keys allow the -1 and -2 rows, and they are smaller and faster to join than text codes.

Special members

- **-1 Unknown:** a reference was missing or could not be matched (for example, a maintenance ticket with no matching vendor code).
- **-2 Not applicable / not yet happened:** a vacant unit-day has no tenant or lease; an open request has no resolved date.

Surrogate-key mapping walkthrough (Lab 05 Task 7.3)

Source payment PAY-0098812 from tenant T-0912, on lease L-2025-00831 for unit GLB01-A101, paid by JazzCash on 14 March 2026 against the rent line of invoice INV-2026-03-00831 (due 10 March):

```
dim_tenant   tenant_code 'T-0912', version valid on 2026-03-14   -> tenant_key 402
dim_unit     unit_nk 'GLB01-A101', version valid on 2026-03-14   -> unit_key 117
dim_lease    lease_number 'L-2025-00831'                         -> lease_key 2210
ownership    owner of GLB01-A101 on 2026-03-14 is 'OW-0217'      -> owner_key 57
dim_payment_method  method_code 'JAZZ'                           -> payment_method_key 5

Resulting fact_payment row:
  paid_date_key 20260314, due_date_key 20260310, unit_key 117, tenant_key 402,
  lease_key 2210, owner_key 57, payment_method_key 5, charge_type_key 1,
  payment_reference 'PAY-0098812', invoice_number 'INV-2026-03-00831',
  invoice_line_no 1 (degenerate dimensions), payment_amount 50000.00,
  days_past_due 4, settlement_lag_days 1, is_reversal FALSE
```

8. Special Dimension Techniques

- **Conformed dimensions.** dim_date, dim_unit, dim_owner, dim_tenant and dim_lease have the same keys and values across facts, which enables drill-across (Q12, Q13).
- **Role-playing dimension.** dim_date is joined under different meanings: invoice, due, paid, snapshot, raised, assigned, resolved, verified, event and coverage date. Time attributes always come from dim_date, never from arithmetic on date_key (Lab 04 Task 5.1).
- **Degenerate dimensions.** invoice_number, invoice_line_no, payment_reference and request_number identify transactions but have no further attributes, so they stay on the fact (Lab 05 Task 5.2).
- **Junk dimension.** dim_request_flags combines three low-cardinality yes/no flags into 8 rows instead of three tiny dimensions.
- **Inferred members.** A fact that arrives before its tenant record links to a placeholder row (is_inferred = TRUE) that is updated when the record arrives.
- **Star, not snowflake.** The location hierarchy lives inside dim_unit, so every question needs one join; a snowflaked Building, City and Region chain was considered and rejected.

9. Attribute Placement Decisions (Lab 05 Task 8.3)

- **Payment method** becomes a small dimension, because Q5 reuses it with a Method → Channel group hierarchy and typical settlement days.
- **Request status** stays on the accumulating snapshot as a degenerate attribute, because it changes at every milestone and belongs to the request.
- **Emergency, recurring and tenant-caused flags** go into a junk dimension, because they have low cardinality and are used together as filters.
- **Owner** is an owner_key on each fact, not an attribute of dim_unit, because ownership changes and the owner at the event date must be kept.

- **Arrears** is a measure (on the occupancy snapshot, and derived from invoices and payments), because a balance is a measurement, not a description of the tenant.
- **Weather** becomes a dimension (city × day), because it is reusable descriptive context and keeps the maintenance fact narrow.

10. Reversals and Credit Notes (Lab 05 Scenario A)

Credit notes are invoice lines with charge type “Credit note” and a negative net_amount. A bounced cheque is a new payment row with a negative amount and is_reversal = TRUE. Original rows are never edited. Reversals share the grain and dimensions of the original transactions, so keeping them in the same fact lets SUM return the correct net figure. A separate returns fact was considered and rejected.

11. Late-Arriving Data

- **Late-arriving dimension.** An invoice for tenant T-0912 arrives before the tenant record. Insert an inferred tenant row and link the fact to it; update that row in place when the real record arrives.
- **Late-arriving fact.** A JazzCash payment made on 2 March arrives on 14 March. Load it at its true paid date, look up the tenant version valid on 2 March, and re-derive the affected occupancy days.
- **Late milestone.** A request resolved last week is marked resolved today. Update the accumulating snapshot row with the true resolved date.